Assignment 01



DOFFING MECHANISM (SPINNING)

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1. Existing Mechanism

The objective of this mechanism is to provide a simple mechanism for simultaneously removing all the filled bobbins and discharging them simultaneously into a suitable receptacle alongside the spinning-frame and also to place on the spindles simultaneously a fresh set of empty bobbins, as more fully hereinafter set forth Doffing mechanisms in spinning machines are crucial for the efficient operation of textile manufacturing. The primary purpose of a doffing mechanism is to remove fully wound yarn packages from spinning machine spindles and replace them with empty bobbins or tubes to continue the spinning process.

The doffing mechanism in spinning machines automates the process of replacing full yarn packages with empty bobbins or tubes. It typically involves a doffer component that rises from below the machine to grip and remove the full packages from the spindles. These full packages are then transferred to a conveyor belt for further processing. Meanwhile, empty bobbins or tubes are loaded onto the spindles to continue the spinning process. This automated system improves efficiency, reduces labor requirements, and ensures smooth operation of the spinning machine.

1.1 Image

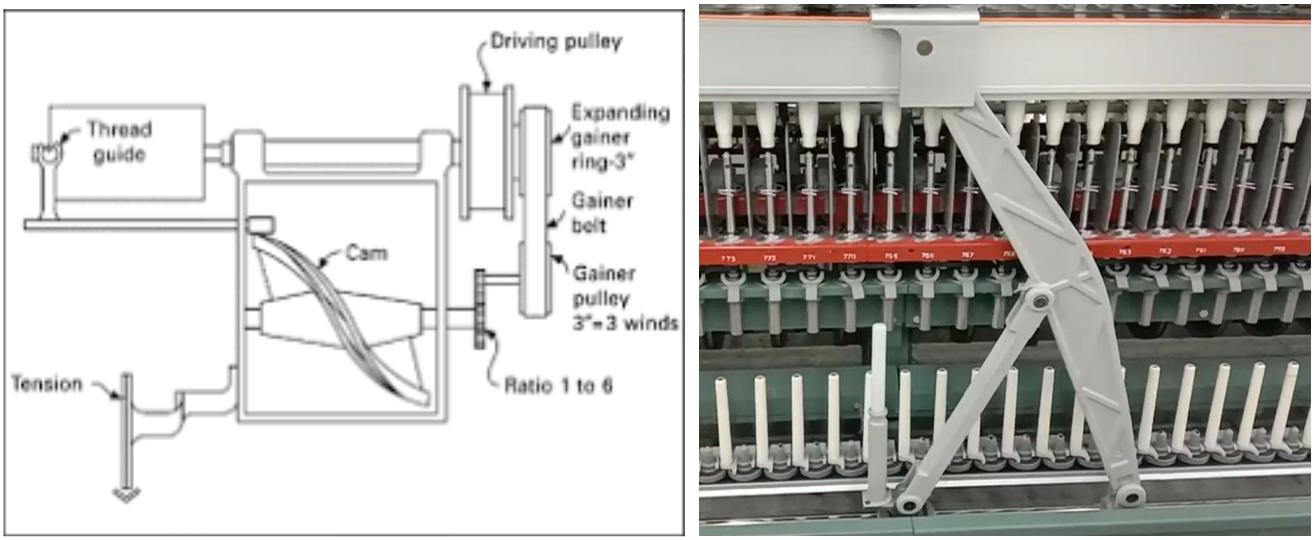


Fig 1: image of doffing mechanism

1.2 Description

When the yarn package reaches the desired size, the spinning machine halts automatically. Following this, the operator switches off the machine and waits for it to come to a complete stop. Afterward, the operator loosens the yarn tail and cuts it with scissors to detach the full yarn

package from the spindle. By grasping the bottom of the package, the operator lifts it up and away from the spindle. Subsequently, the operator replaces the empty tube or bobbin on the spindle, securing it in place. With the spinning machine restarted, the operator initiates the process of spinning a new yarn package.
There are two main types of doffing mechanisms:
● Automatic Doffing: These mechanisms are automated and perform the doffing process without the need for manual intervention. Automatic doffing mechanisms can be further classified into stationary and traveling devices, each with its own method of operation.
● Manual Doffing: In manual doffing, operators manually remove the full cops from the spinning machine and replace them with empty bobbins or tubes. This process is labor-intensive and less efficient compared to automatic doffing.

Doffing mechanisms typically consist of components such as:
● Doffer: The doffer is the main component responsible for removing full cops from the spindles and transferring them to a conveyor belt or other collection area. It may also hold empty bobbins or tubes for reloading onto the spindles.
● Conveyor Belt: A conveyor belt is often used to transport the full cops
away from the spinning machine to a winding machine or another processing station.
● Actuators and Sensors: These components help control the movement and timing of the doffing process, ensuring smooth operation and preventing errors or jams. In automatic doffing mechanisms, the process typically involves the following steps:
● The doffer rises from below the spinning machine, positioning itself to access the full cops on the spindles.
● Fully wounded cops are gripped by the doffer and transferred to it, while empty bobbins or tubes are loaded onto the spindles.
● Once all full cops have been removed from the spindles, the doffer returns to its original position.
● The full cops are then transferred to a conveyor belt for further processing or packaging
Expected velocity as per the analysis done is 12rpm.

1.2.1 working principle

The working principle of the doffing mechanism in spinning involves the following steps:

1. Detection of Full Bobbins: Sensors detect when bobbins are full and ready for removal.
2. Grasping and Removal: A grasping unit secures the full bobbin and removes it from the spindle.
3. Replacement with Empty Bobbins: The mechanism aligns and places empty bobbins onto the spinning spindles.

4. **Thread Management**: Threads are severed cleanly to prevent tangling or breakage during the replacement process.
5. **Automation and Movement**: The doffing apparatus moves along the spinning frame, performing these operations efficiently to minimize downtime.

1.2.1 Expected speed

With a cycle time of **180 seconds**, the RPM is approximately **0.33 RPM**.

2. Kinematic sketch

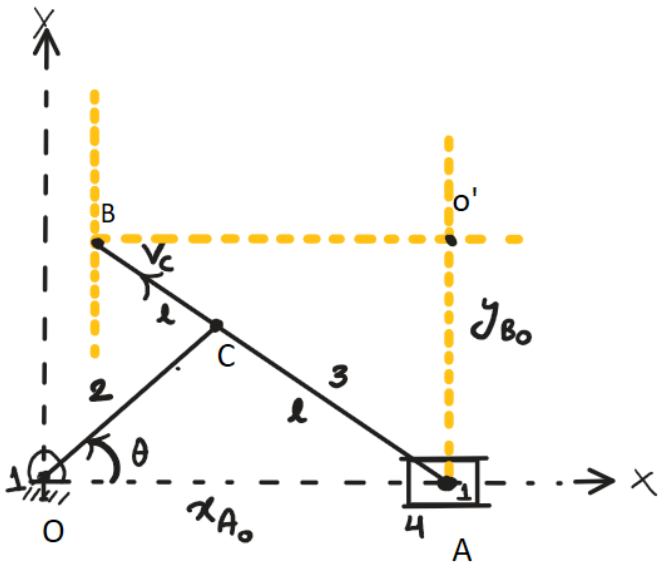


Fig.2 Kinematic sketch of mechanism

3. Simulation of Original Mechanism in Software

The dimensions of the mechanism are determined according to the specifications of the machine and the desired output at the focal point. Additionally, angular and linear speeds are estimated based on the range of operating conditions.

These are the isometric and front view of the approximate mechanisms modeled in SOLIDWORKS software. Here the point P on the connecting link is of our interest and hence referred to as the output link and all further analysis is carried out for this particular link and point only.

Table 1

Link No.	Name	Link type
Link 1	Ground	Fixed
Link 2	Crank	Supporting
Link 3	Connecting rod	Output
Link 4	Slider	Input

Table 2

Connection	Joint Type
Ground and crank	Revolute
Crank and coupler	Revolute
Coupler and slider	Revolute
Slider and ground	Translation

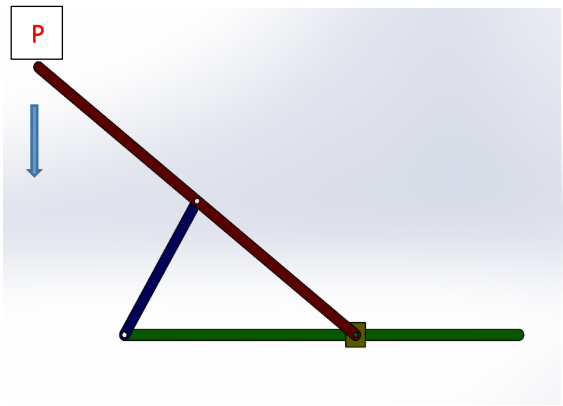


Fig.3 Orthographic view of mechanism

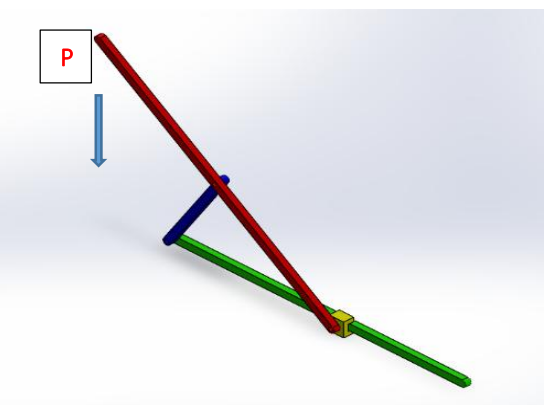


Fig.4 Isometric view of Mechanism

4. Kinematic Analysis of Original Mechanism: Simulation in Software

4.1 Displacement curves

4.1.1 Displacement:

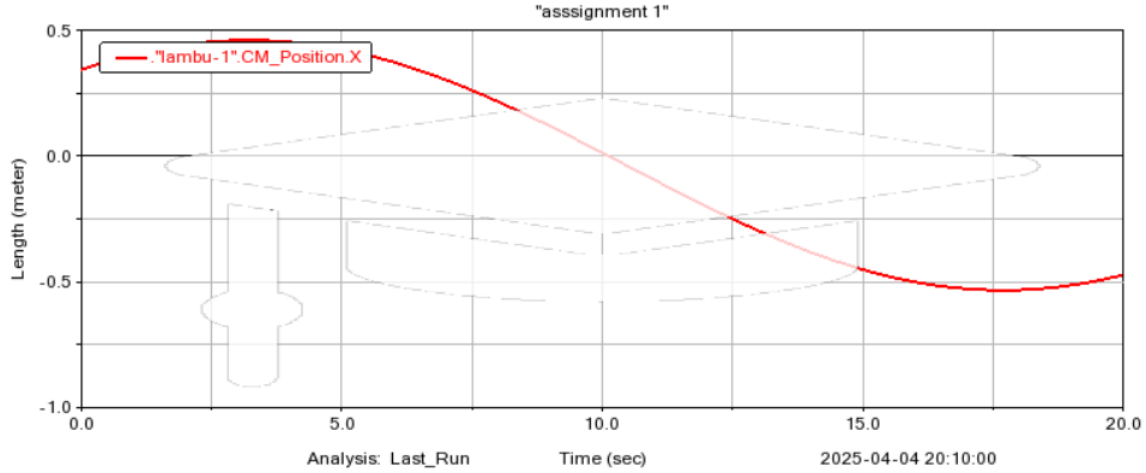


Fig.5 Displacement curve x direction

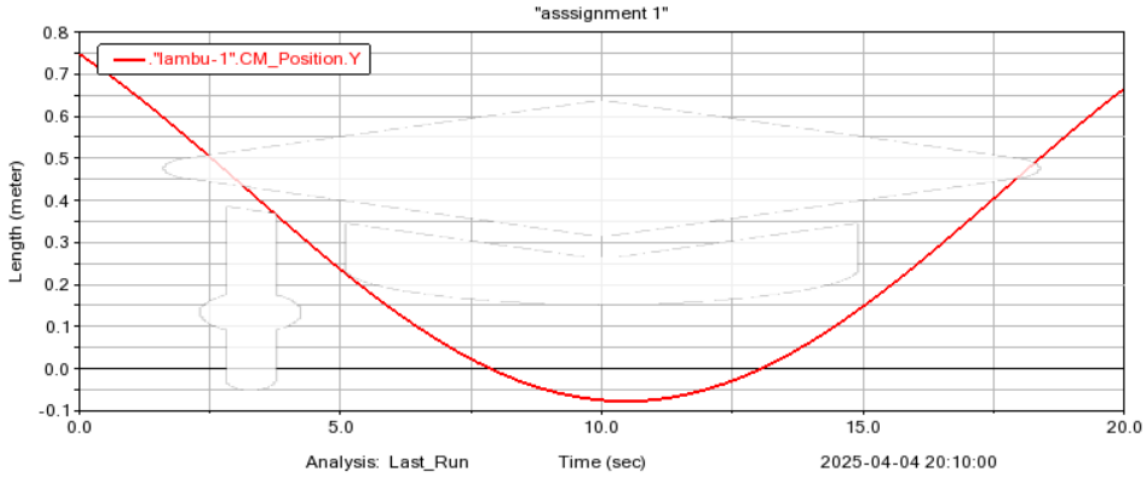


Fig.6 Displacement curve Y direction

4.1.2 Trace:

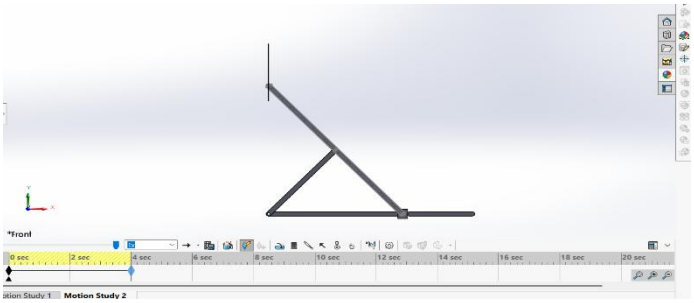


Fig.7 the y component of displacement vs. x component of displacement.

4.2 Velocity curve

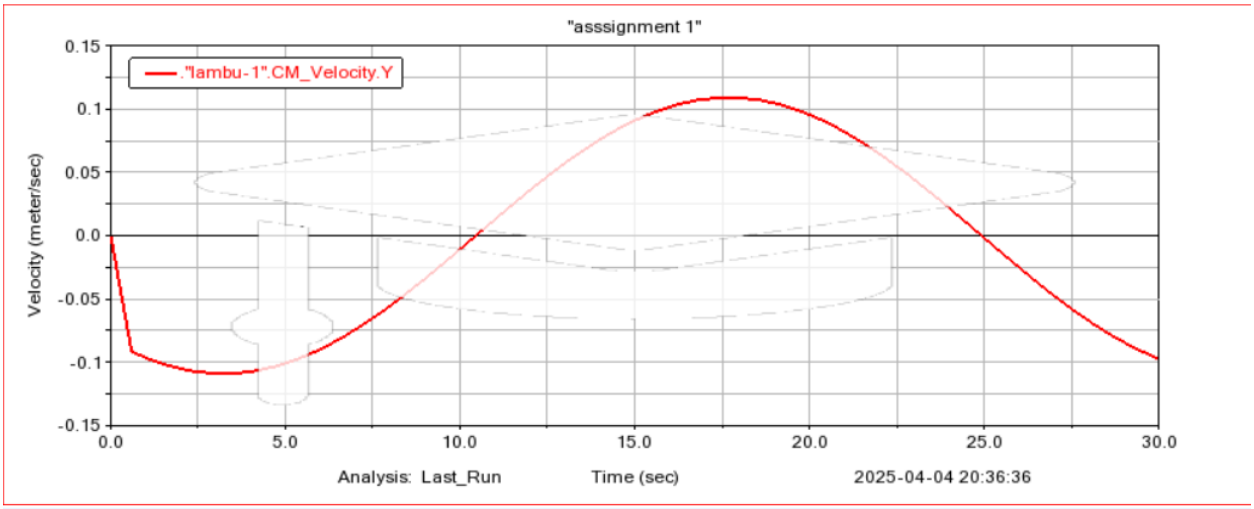


Fig.8 velocity curve

4.3 Acceleration curve

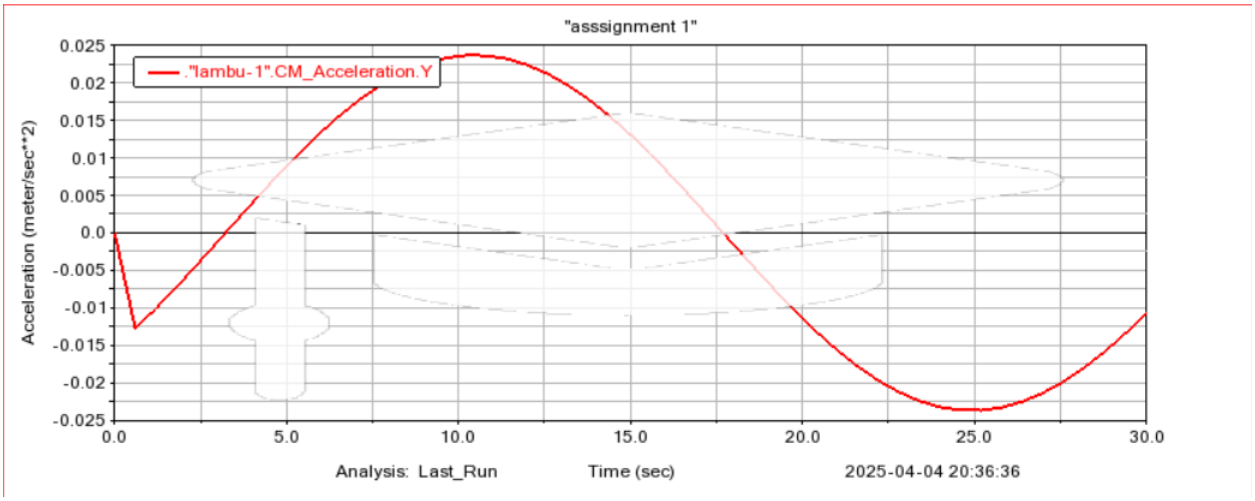


Fig.8 Acceleration curve

4.4 Comments

1. Displacement : The crank-slider mechanism allows for controlled linear motion of the chosen point along a specified path. By varying the crank radius and connecting rod length, the displacement of the point can be precisely adjusted to meet the requirements of the application.

2. Velocity : Through kinematic analysis, the velocity profile of the chosen point in the crank-slider mechanism can be determined. The velocity varies sinusoidal with respect to crank angle, reaching maximum and minimum values at certain points in the cycle. Understanding this velocity profile is crucial for optimizing the motion characteristics of the mechanism.

3. **Acceleration** : Dynamic analysis of the crank-slider mechanism reveals the acceleration profile of the chosen point throughout its motion. This information is essential for assessing the forces and loads experienced by the mechanism components, as well as ensuring smooth operation and minimizing vibration or impact forces.

5. Dynamic Analysis of Original Mechanism : Simulation in Software

5.1 Total mass

Link	Mass (KG)
Ground	3.93
Crank	1.98
Connecting rod	3.93
Slider	0.54

The total mass of the mechanism is 10.38 kg

5.2 Torque/Force

For the dynamic analysis the force of 10 N is applied on the output point vertically and the torque is calculated in input joint.

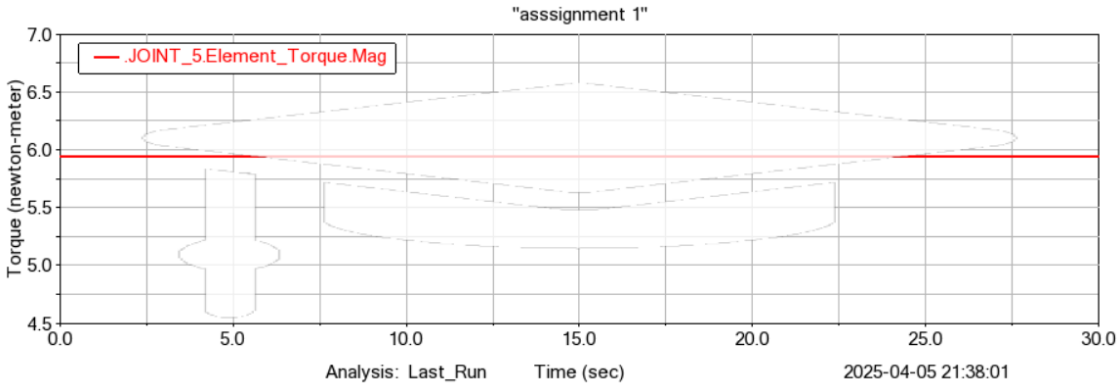


Fig.10 Torque curve

5.3 Power

The power required by the input motor can be easily obtained by considering the average motor torque(6N-m) required and then multiplying by the constant angular velocity of the input arm which we assume to be N = 5 rpm

Power= T * **ω**= 6*0.523= 3.1412 Watt

6. Alternate Linkage

An attempt was made to devise a new linkage that would perform the same task and to get similar output. Images attached below is the kinematic linkage sketch of the proposed alternate mechanism with only lower pairs showing different links. This is a simple slider crank mechanism in which slider exactly duplicate the motion of connecting rod point P at the original mechanism.

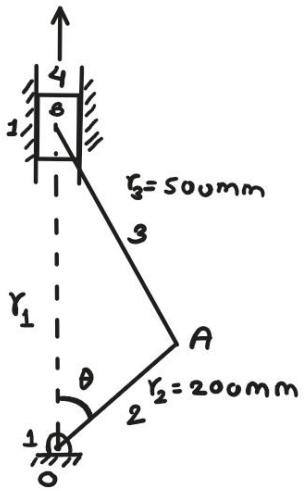


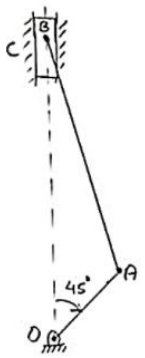
Fig 11: kinematic sketch of alternate mechanism

7 Kinematic Analysis of Alternate Linkage: Manual Calculation

7.1 velocity analysis

The crank of the alternate mechanism is rotated at the speed of 5 rpm and velocity of the slider at 45^0 position of crank and Output is chosen as slider Velocity analysis is done graphically as follow:

Scale 100mm = 10mm



Velocity analysis

N = 5RPM ω = 0.523 rad/sec

$V = \omega r = 0.1047$ m/sec

Scale = 50mm = 0.1047 m/sec

1mm = 2.094×10^{-3} m/sec

$V_{AB} = ab = 37\text{mm} = 0.077$ m/sec

$V_{BC} = bc = 46\text{mm} = 0.096$ m/sec

$V_{slider} = 96.32$ mm/sec

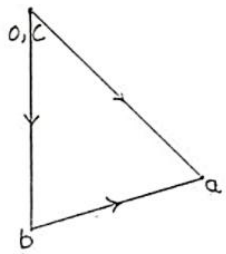


fig: velocity polygon

Velocity of the output link (slider) I found to be 96.32mm/sec

7.2 Acceleration analysis

Acceleration analysis is done graphically as follow:

acceleration analysis

Scale = 0.0548 = 50 mm 1 m/sec^2 = 912.4 mm

$a_{OA}^{centripetal} = \frac{V_{OA}^2}{r_{OA}} = 0.0548$ m/sec^2

$a_{OA}^{tangential} = 0$

$a_{AB}^{con.} = \frac{V_{AB}^2}{r_{AB}} = 0.0118$ m/sec^2 = 10.766 mm

a_{BC} = 33 mm = 0.0361 m/sec^2 = 36.168 mm/sec^2

acceleration of slider = 36.168 mm/sec^2.

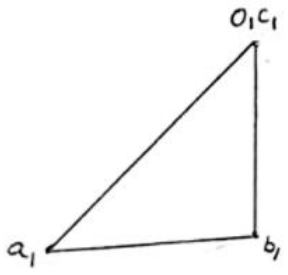


fig - acceleration polygon

Acceleration of output is found to be 36.168mm/sec^2

8. Kinematic Analysis of Alternate Linkage: Simulation in Software

8.1 Modeling the alternate mechanism

8.1.1

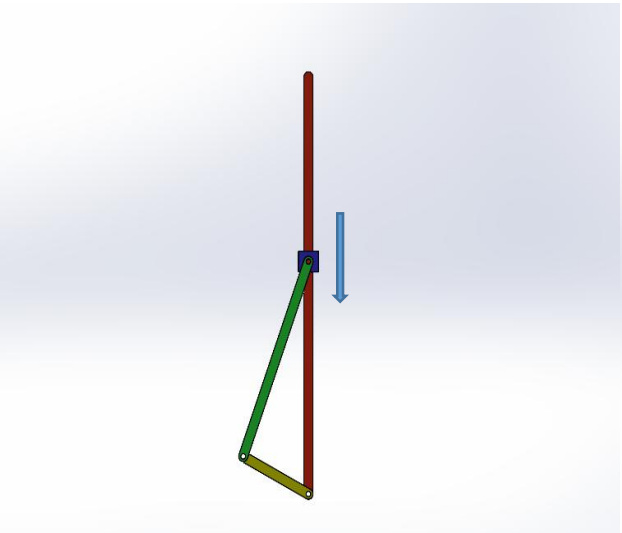


Fig.12 Orthographic view of Alternate mechanism

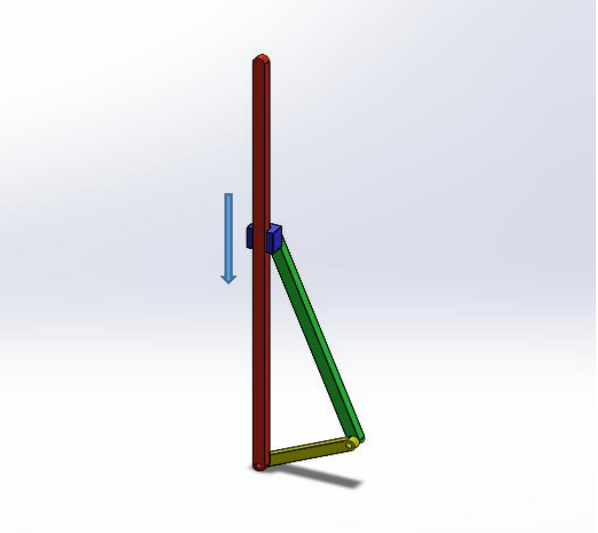


Fig.13 Isometric view of Alternate Mechanism

8.2 Displacement

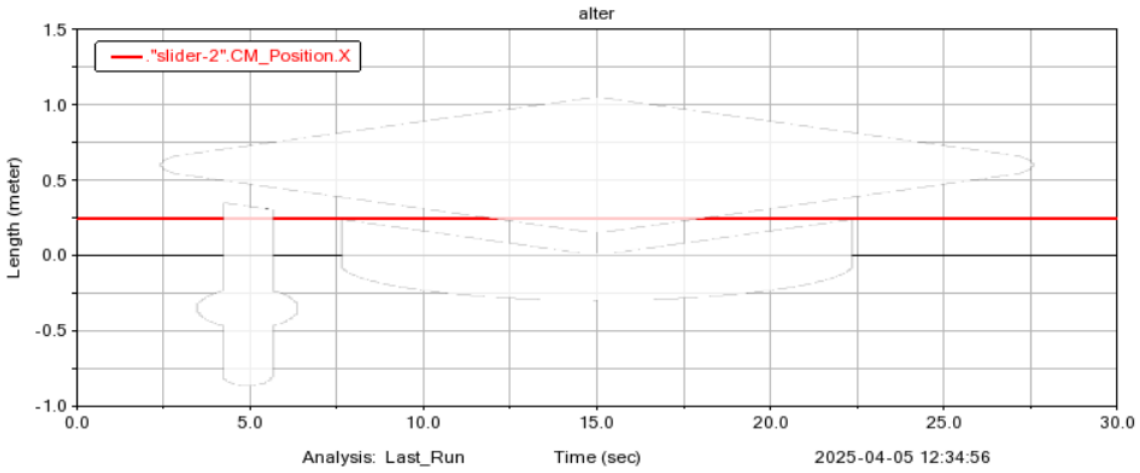


Fig.14 displacement (X-axis) curve Alternate Mechanism

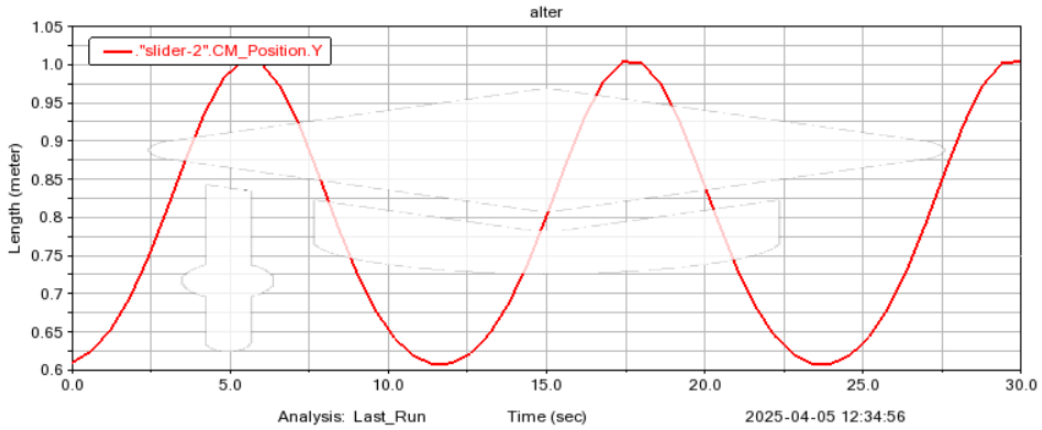


Fig.15 displacement (Y-axis) curve Alternate Mechanism

8.3 trace

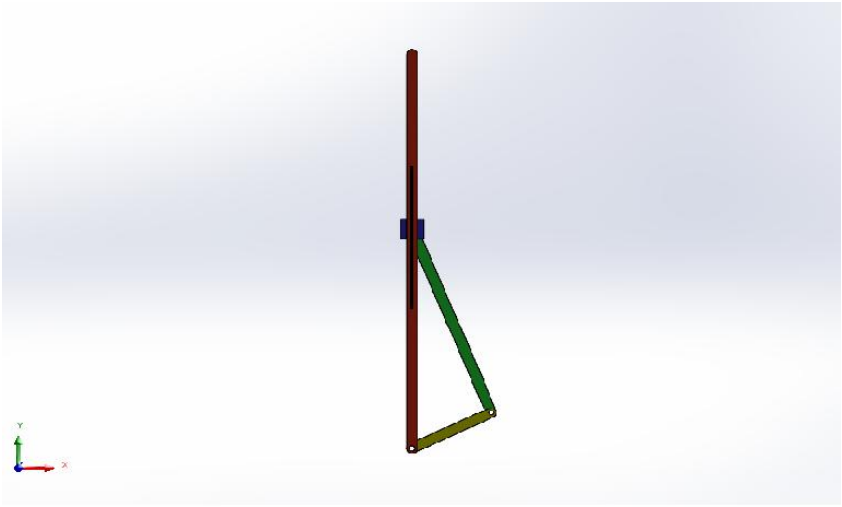


Fig.16 Trace curve Alternate Mechanism

Here the black color line in the figure indicate the trace of the mechanism

9. Dynamic Analysis of Alternate Linkage: Simulation in Software

9.1 Total Mass

link	Mass(KG)
Crank	0.81
Connecting rod	1.98
slider	0.54

Total mass of the mechanism is 3.33 kg

9.2 Torque/Force

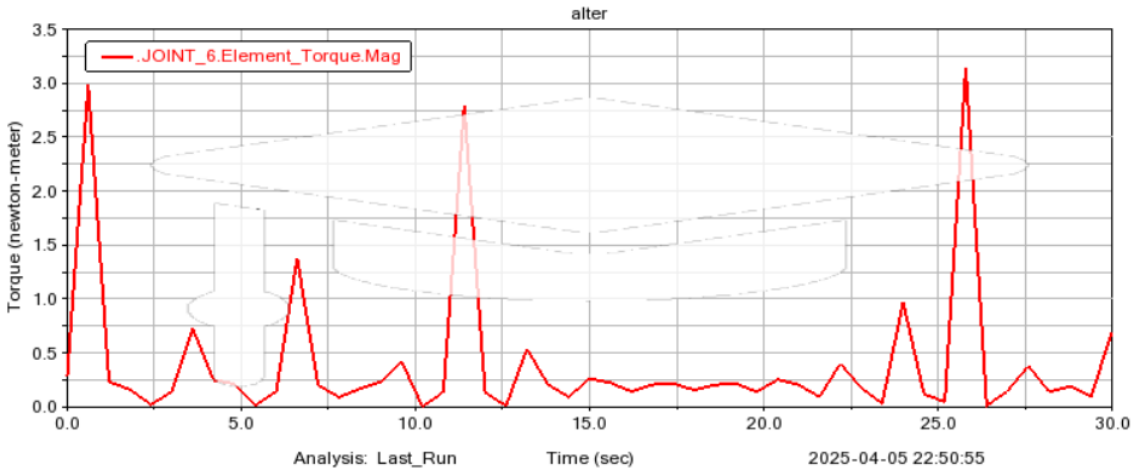


Fig.18 Torque curve Alternate Mechanism

9.3 Power

The power required by the input motor can be easily obtained by considering the average motor torque(3N-m) required and then multiplying by the constant angular velocity of the input arm which comes out to be Speed of the input crank N = 5 rpm.

Power= T * **ω**= 3*0.523= 1.57 Watt

10 Comparison of Original and Alternate Mechanism

10.1 Displacement Comparison

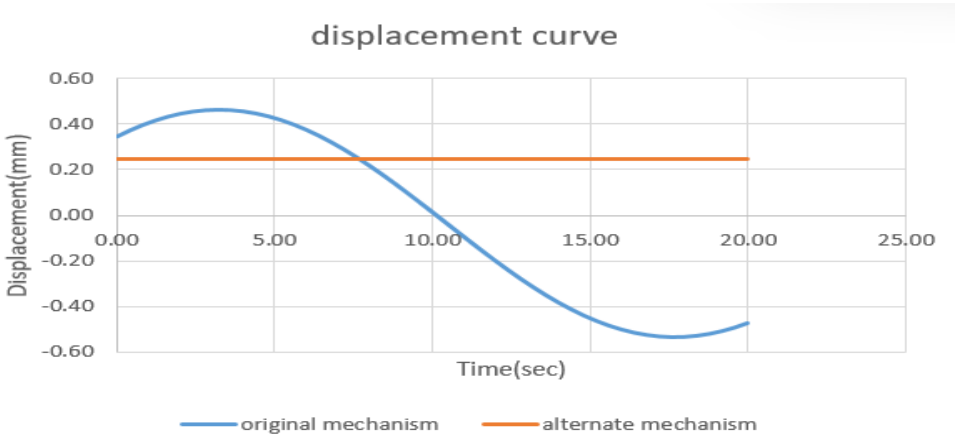


Fig.19 Displacement Comparison x-axis

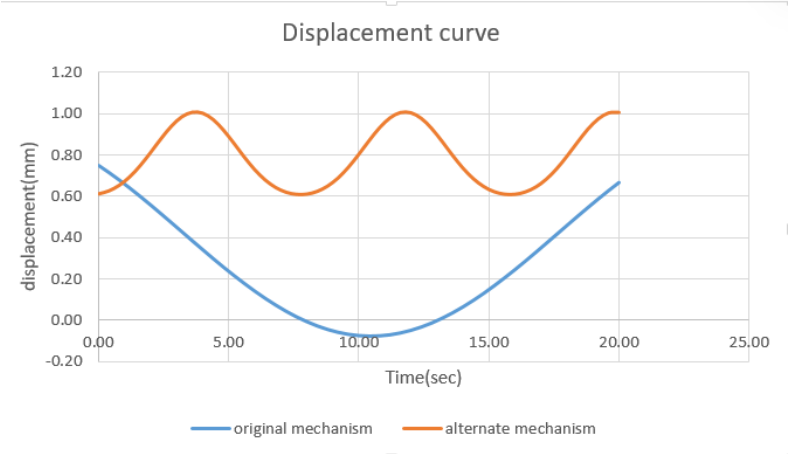


Fig.20 Displacement Comparison y axis

10.2 Trace Comparison:

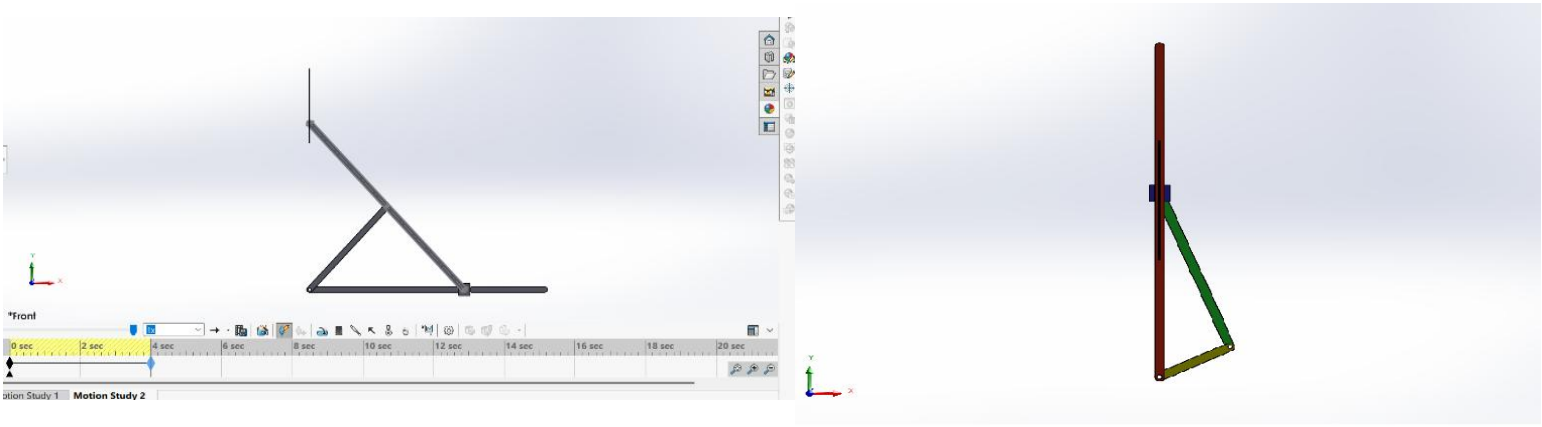


Fig.21 Trace Comparison between two mechanisms

Both of the traces are vertical straight line as required and also same as shown in the above figure.

10.3 Torque/Force Comparison

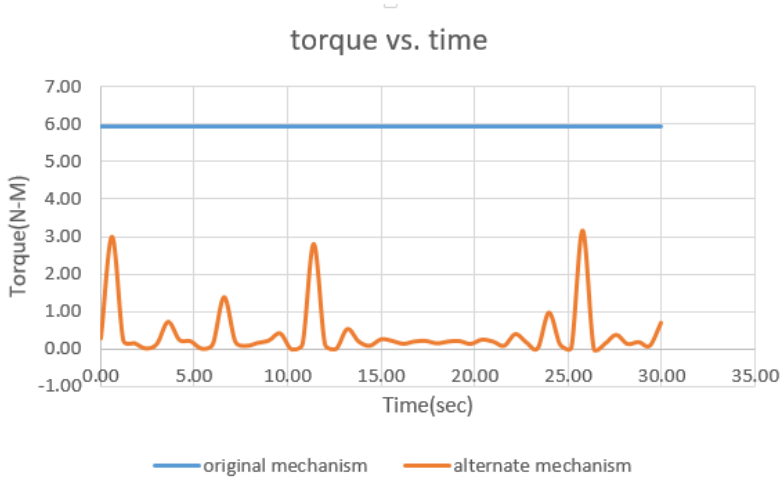


Fig.22 Torque Comparison

Torque has significance difference in original and alternate mechanism as shown in the fig.22

10.4 Power Comparison:

Power to run original mechanism is found to be 3.1415 watt and for alternate mechanism is 1.57 watt i.e. alternate mechanism run at lower power.

11. Comments

11.1 Error Analysis

From the above displacement comparison it is clearly shown deviation in both the axis where these may occur due to the input speed during kinematic analysis because I take very low speed at input arm which is the machine operating speed and for alternate mechanism I guess the speed to be an order higher for kinematic analysis. Error can also may occur due to the step size taken during the simulation.

11.2 Index of Merit

The alternate mechanism exhibits superior performance compared to the original in my case.

- The alternative mechanism having a less mass than existing one.
- the requisite dwell motion, critical for functionality, is insufficient in the alternative mechanism.
- the alternative mechanism demands a lower power input than the existing one.
- Consequently, the alternate mechanism demonstrates a superior index of merit in technical evaluation.

References:

- *https://www.sciencedirect.com/science/article/abs/pii/B9781845694449500073*
- *https://patents.google.com/patent/US1608912*
- ***https://www.youtube.com/watch?v=FXPONEEcSkI***